

Smart Water Pitcher Base

[CSE123]

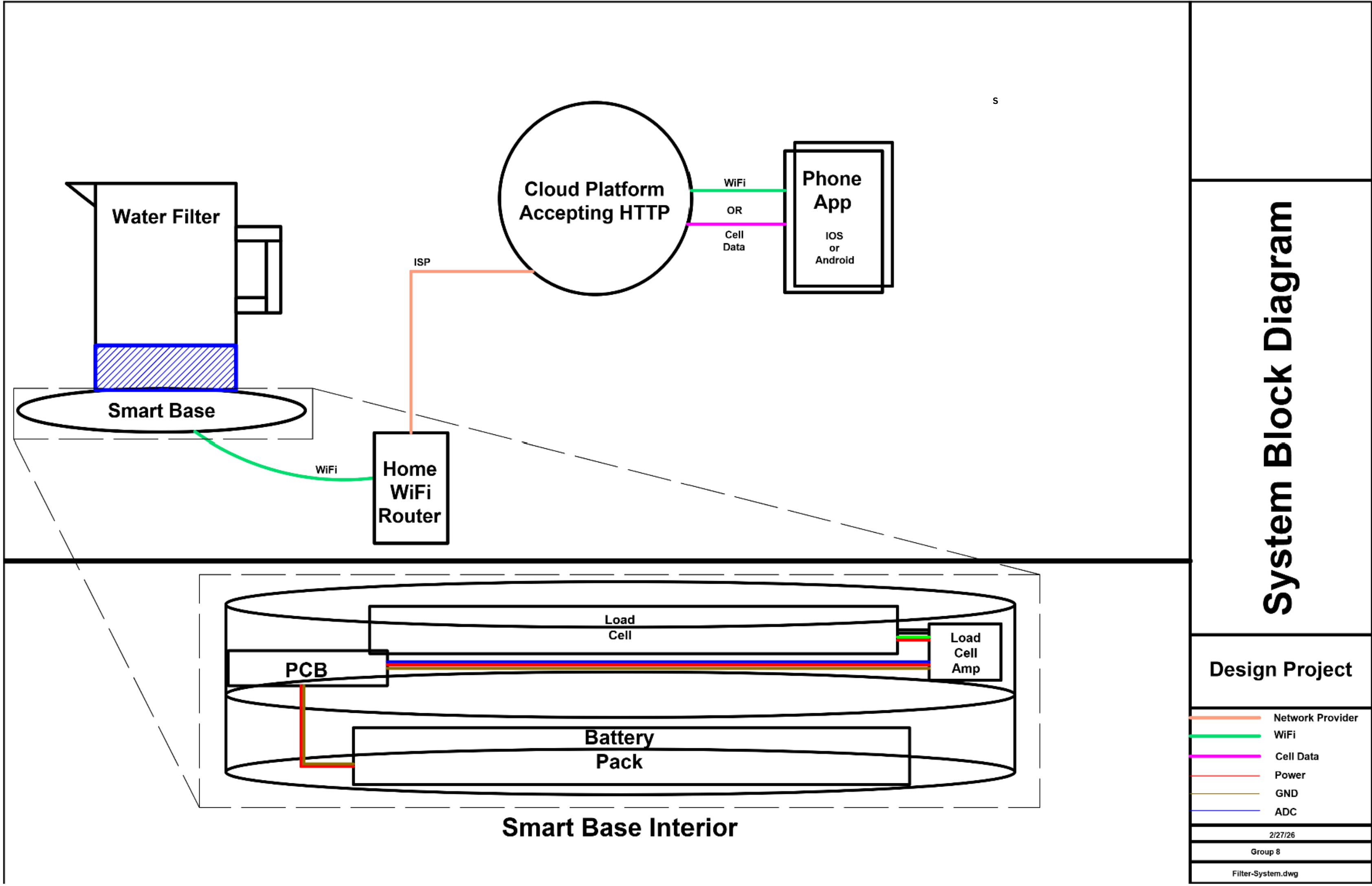


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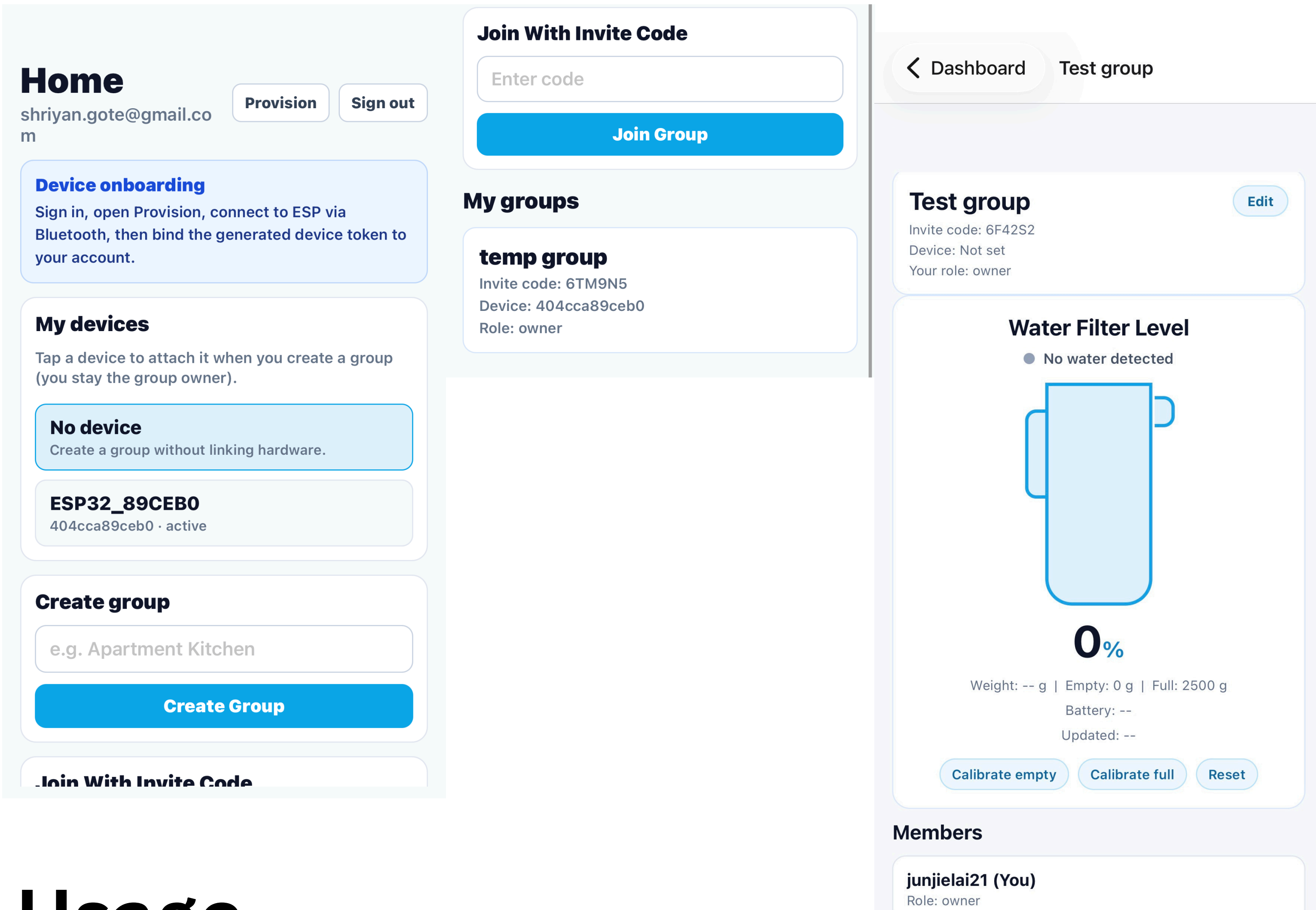
Introduction

Living with roommates can sometimes mean your water filter device is empty when you need water. So, we designed a device to know how full your water pitcher is at any moment of the day. The design is intended to keep all users up to date with the status, as well as inform users if the level goes below a threshold.

System Overview



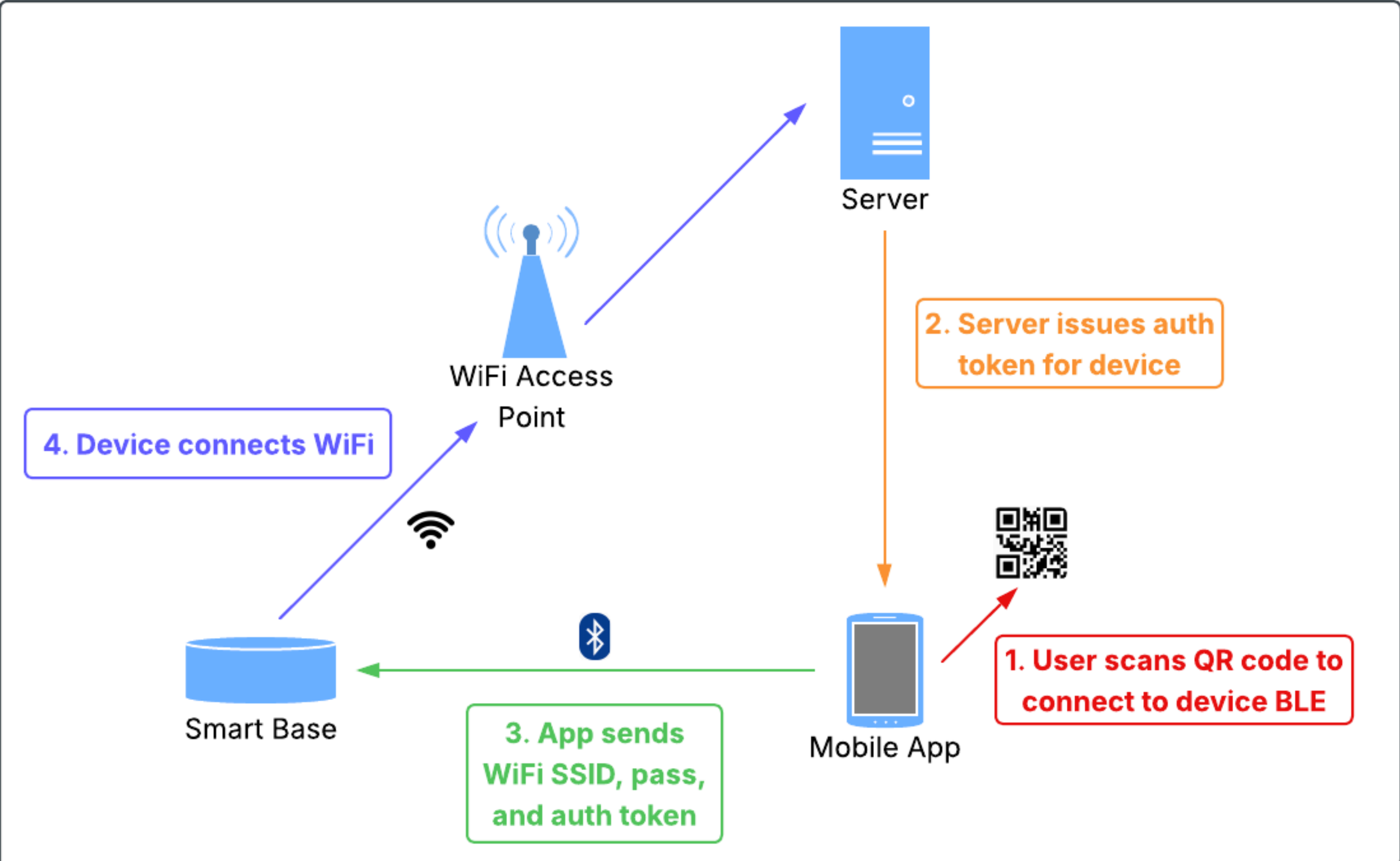
Mobile App



Usage

1. Connect smart base to WiFi network

Smart Base WiFi Provisioning



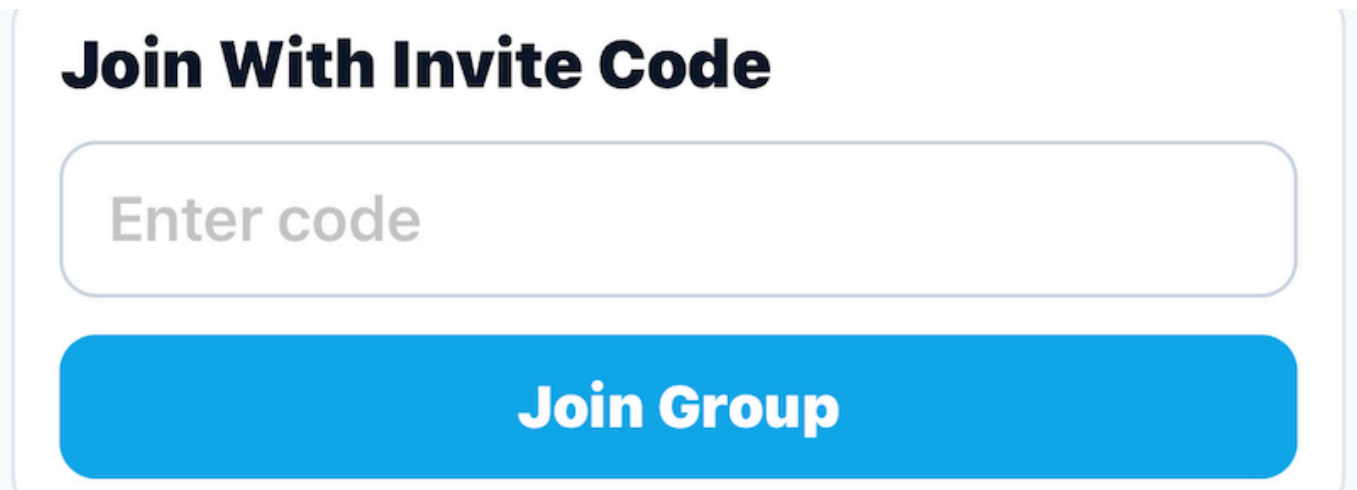
2. Calibrate base

Calibrate empty

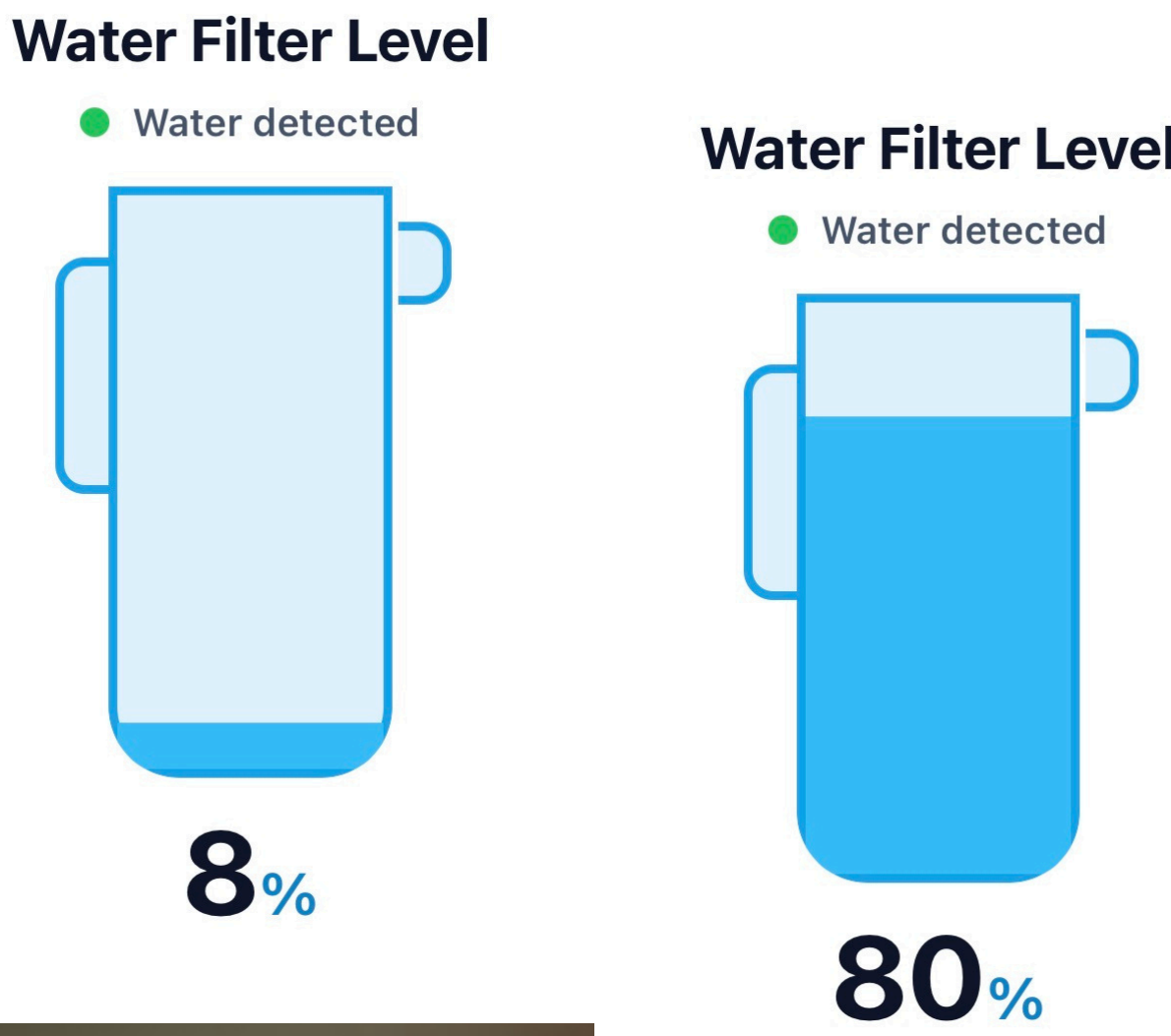
Calibrate full

Reset

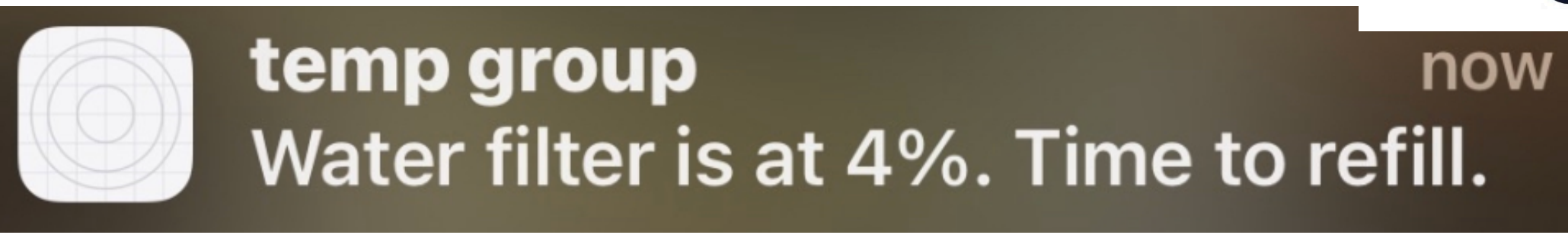
3. Join a group on the mobile app



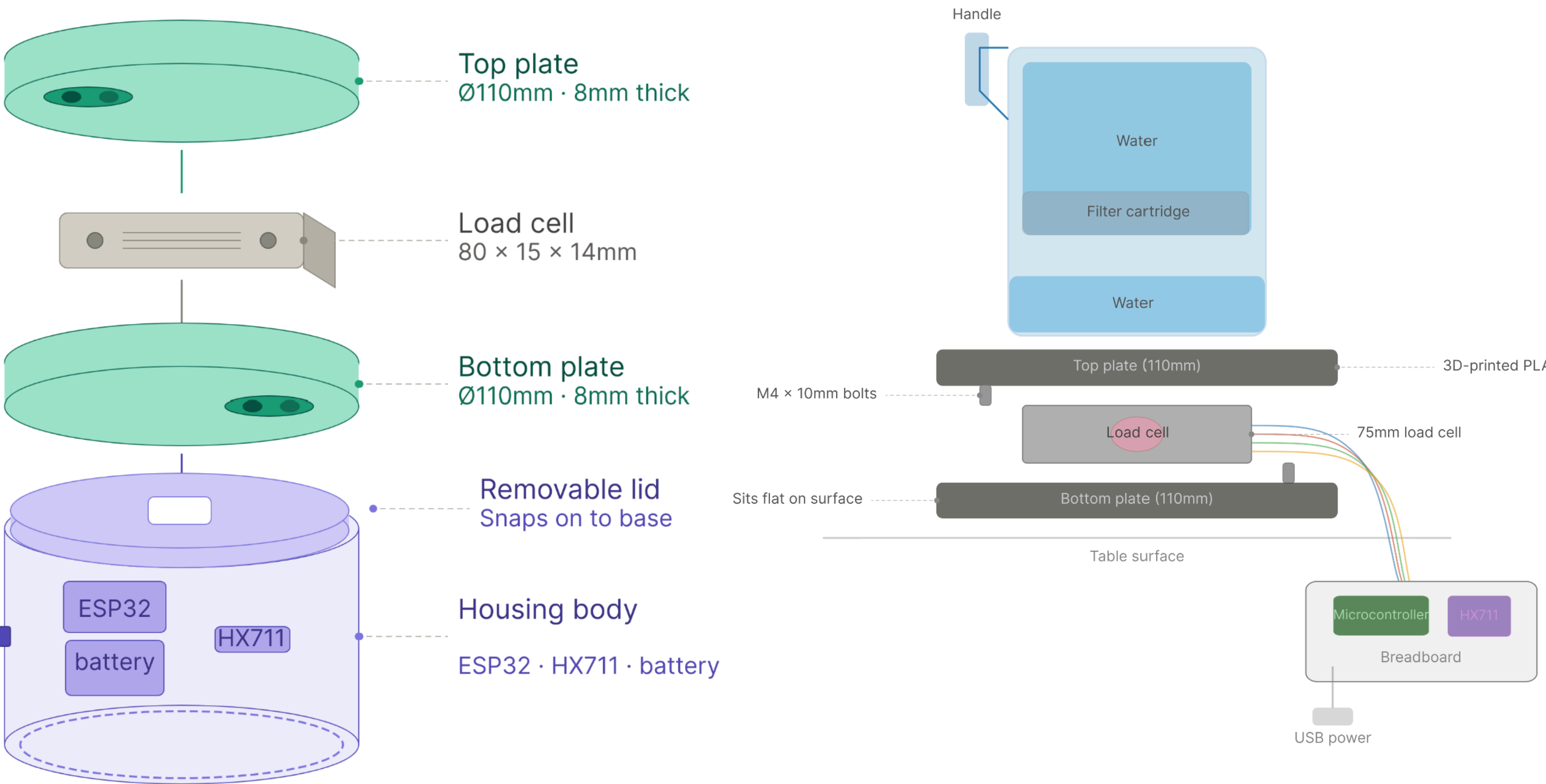
4. Check water level at any time



5. Receive notifications when water



Smart Base



Smart Base:

- ESP32C3 microcontroller
- HX711 amplifier
- 3D printed structure

Mobile App & Server:

- React Native
- Vercel webserver
- Supabase database